

FH Aachen

Faculty of Aerospace Engineering

Space Flight Mechanics

Bachelor Thesis

**Usage of Aerobraking
in the Saturnian System**

Kalscheuer, Jason Cedric

Aachen, December 18, 2024

Examiner: Dachwald, Bernd, Prof. Dr.-Ing.

Second Examiner:

Contents

List of Figures	3
List of Tables	4
1 Introduction	5
2 Description of the program code	6
2.1 User Input	6
2.2 Preparing the program	8
2.3 Atmospheric Model	9
2.4 Mission phases	9
2.5 Equations of Motion	9
2.6 Launch Delay Calculation	12
2.7 Adjust Aerobrake Altitude	12
3 Results	13
3.1 Simulation results	13
3.2 Validation of results	16
3.2.1 Validation of the atmospheric model	16
3.2.2 Validation of aerobraking results	16
4 Conclusion	17
A Appendix Title	18
A.1 A First Section in the Appendix	18

List of Figures

2.1	Atmospheric Density Model	9
3.1	Heating Rate	13
3.2	Heating Rate Legend	14
3.3	Trajectory Example	14
3.4	Trajectory Example detailed	15
3.5	Possible First Crossing Distances based on Arrival velocity	15
3.6	Atmospheric Density Data from HASI	16

List of Tables

2.1	Rocketparameters (own figure)	6
2.2	Orbital parameters (own figure)	7
2.3	Outputparameters (own figure)	7
2.4	Planetary and Orbital constants (own figure)	8
2.5	Outputparameters (own figure)	10

1 Introduction

Please add your introductory text here.

2 Description of the program code

2.1 User Input

The user input is the first region of the program and is used to modify the simulation to be as close to the planned mission as possible. It is divided into the three categories Rocket parameters, Orbital parameters, and Output parameters. This part begins with "region User input" and ends with "endregion".

Constant	Name in program	Value	Unit
Total mass	m_0	3725.0	kg
Propellant mass	m_p	2725.0	kg
Thrust	thrust	0	N
Hyperbolic excess velocity	v_inf	1 .. 3	km/s
Specific Impulse	Isp	3100	s
Drag Coefficient	c_d	2.2	–
Cross-Sectional Area	A	29.3	m ²
Heat Load Coefficient	c_q	0.5	–

Table 2.1: Rocketparameters (own figure)

Constant	Name in program	Value	Unit
Simulation duration	tint	540	days
Lowest periapsis	r_p_lowest	258	km
Highest periapsis	r_p_highest	260	km
Step size	r_p_step_size	0.1	km
Periapsis of desired Orbit	r_p_orbit	1500	km
Apoapsis of desired Orbit	r_a	2000	km

Table 2.2: Orbital parameters (own figure)

Constant	Name in program	Options
Simulation duration	Plot_Trajectory	True/False
Lowest periapsis	Plot_Atmosphere	True/False
Highest periapsis	Plot_Values	True/False
Step size	Optimize_Thrust_Duration	True/False
Periapsis of desired Orbit	Plot_Heating_Rate	True/False

Table 2.3: Outputparameters (own figure)

2.2 Preparing the program

This section of the code begins with "region Prepare the program" and ends with "endregion". It consists of general planetary and orbital constants, accuracy parameters for the simulation, Unit conversion as well as initializing variables to track the spacecrafts state during the simulation.

Constant	Name in program	Value	Unit
Gravitational Acceleration	g0_earth	9.80665	m/s ²
Gravitational Parameter of Saturn	mu_saturn	37931207.8	km ³ /s ²
Radius of Saturn	R_saturn	60268.0	km
Distance between Saturn and Titan	dist_saturn_titan	122187.0	km
Saturn's Sphere of Influence Radius	R_SOI_Saturn	54500000	km
Gravitational Parameter of Titan	mu_titan	8978.14	km ³ /s ²
Radius of Titan	R_titan	2574.73	km
Titan's Sphere of Influence Radius	R_SOI_titan	1200000	km
Titan's semi-major axis	a_titan	1221870	km
Eccentricity of Titan's Orbit	e_titan	0.0288	–
Orbital Period of Titan	T_titan	15.945	days
Gravitational Parameter of Enceladus	mu_titan	7202	km ³ /s ²
Radius of Enceladus	R_titan	252.1	km
Enceladus' semi-major axis	a_titan	237948	km
Eccentricity of Enceladus' Orbit	e_titan	0.0047	–
Orbital Period of Enceladus	T_titan	1.370218	days

Table 2.4: Planetary and Orbital constants (own figure)

2.3 Atmospheric Model

The atmospheric model is a standard exponential model and extends up to 1375km altitude. The base value at the surface level is 5.38 kg/m^3 . At its farthest point, this value decreases to $1\text{e-}11 \text{ kg/m}^3$. The atmospheric model is based on the data from the Huygens Atmospheric Structure Instrument. A more in-depth comparison is found in 3.2.1 (VERLINKEN)

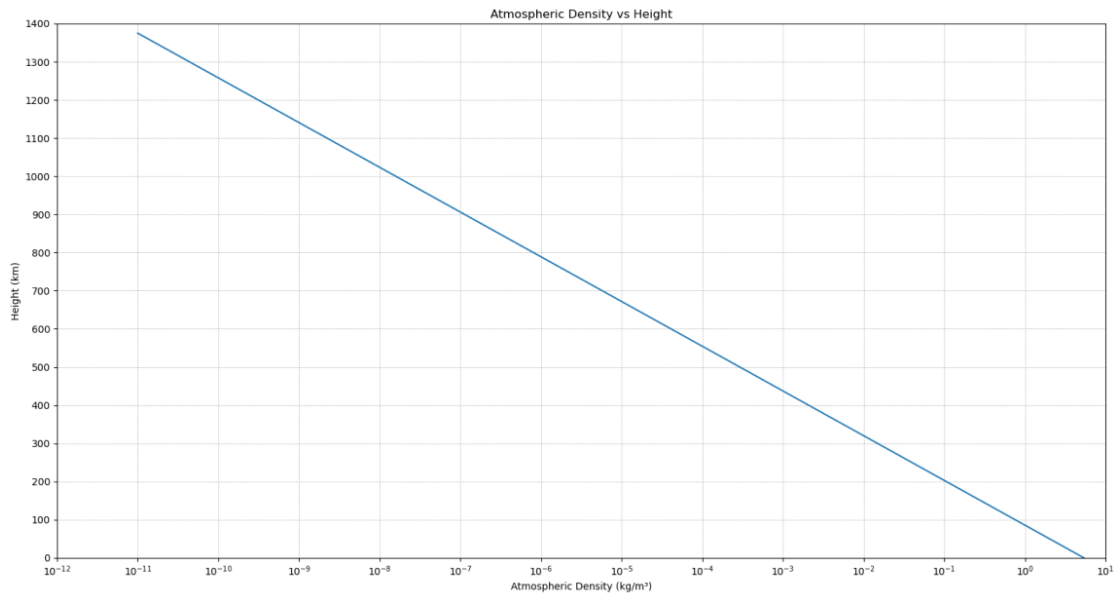


Figure 2.1: Atmospheric Density Model

2.4 Mission phases

2.5 Equations of Motion

The equations of motion section of the code contains four functions. The three helper functions "get_titan_position", "get_enceladus_position", "get_initial_spacecraft_state" and the main function that calculates the following parameters of the spacecraft for each point in the simulation.

Parameter	Name in program	Unit
Velocity	v	km/s
Flight path angle	alpha	rad
Specific orbital energy	E	MJ/kg
Specific angular momentum	h	km/s ²
semi-major axis	a	km
eccentricity	e	–
apoapsis	apoapsis	km
periapsis	periapsis	km
descending	is_descending	True/False
throttle	throttle	0/1
atmospheric density	atmo_density	kg/km ³
Airdrag	D	kg * km/s ²

Table 2.5: Outputparameters (own figure)

The following five differential equations are solved in the main function and determine the spacecraft status parameters listed above in Table 2.5 at each simulation time step.

$$\frac{dr}{dt} = \rho_r \quad (2.1)$$

$$\frac{d\phi}{dt} = \rho_\phi \quad (2.2)$$

$$\frac{d\rho_r}{dt} = r \rho_\phi^2 - \frac{\mu}{r^2} - \frac{D}{m} \cos(\alpha) + \frac{\text{throttle} \cdot \text{thrust}}{m} \cos(\beta) \quad (2.3)$$

$$\frac{d\rho_\phi}{dt} = -2 \rho_r \rho_\phi - \frac{D}{m} \sin(\alpha) + \frac{\text{throttle} \cdot \text{thrust}}{m} \sin(\beta) \quad (2.4)$$

$$\frac{dm}{dt} = - \frac{\text{throttle} \cdot \text{thrust}}{c_{\text{eff}}} \quad (2.5)$$

2.6 Launch Delay Calculation

The Launch Delay Calculation simulates the trajectory of the spacecraft until it crosses Titans Orbit the first time and checks for the angular difference between the spacecraft and Titan at that time. It aims to find a launch delay that results in the spacecraft passing in front of Titan at the distance given in the User Input region of the program. It utilizes a database as starting values made from delay values that have proven to result in fast convergence of the iteration process. The iteration process follows a distinct logic of checking for the angular difference first and adjusting the delay according to that. If the angular difference is positive, since we plan to pass in front of Titan and 2° the adjustment will be based on the error to the target crossing distance. Every adjustment also includes the hyperbolic excess velocity as a factor. When a suitable delay time is found, the program moves to the next section to adjust the Aerobrake Altitude.

2.7 Adjust Aerobrake Altitude

This function utilizes the launch delay that was found by the Launch Delay Calculation. It then runs a longer simulation with that delay and checks for the second crossing of Titan Orbit and checks for the angular difference between the spacecraft and Titan.

3 Results

Analyzing possible aerobraking altitudes to stay within the Saturnian system results in finding multiple possible heights for every hyperbolic excess velocity up to 3 km/s. Higher aerobraking altitudes reduce the speed of the spacecraft less and therefore result in a longer time until the next aerobraking maneuver and a longer mission duration overall. However, there is an advantage in another area, because the heat load on the spacecraft at heights above 600 km for the first crossing is significantly lower. In fact, low enough so that no extra heat shielding is necessary and radiative heat dissipation using coatings is sufficient.

3.1 Simulation results

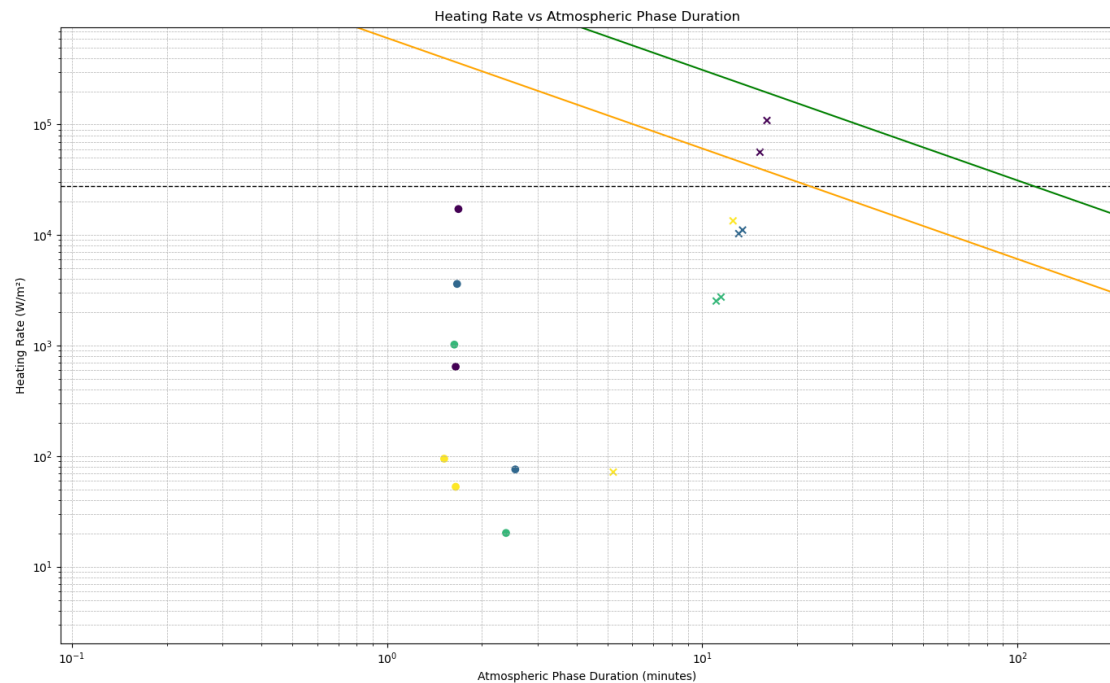


Figure 3.1: Heating Rate

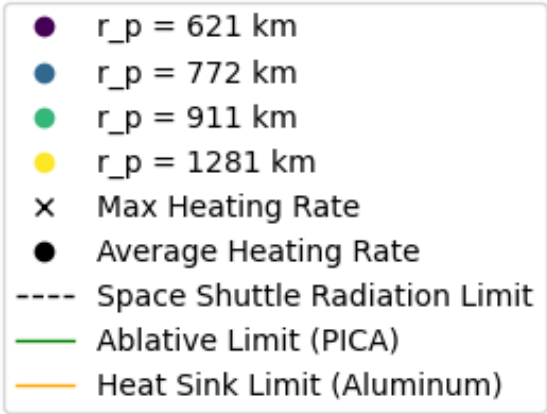


Figure 3.2: Heating Rate Legend

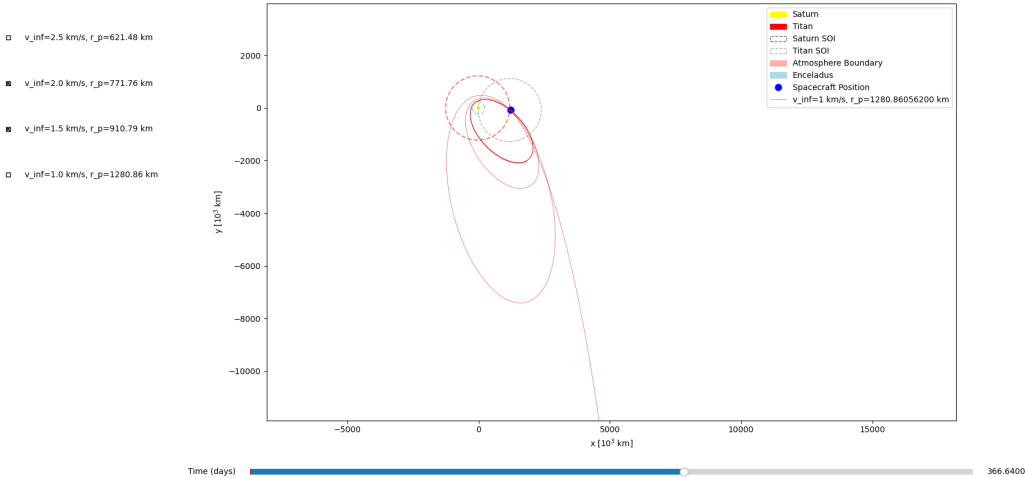


Figure 3.3: Trajectory Example

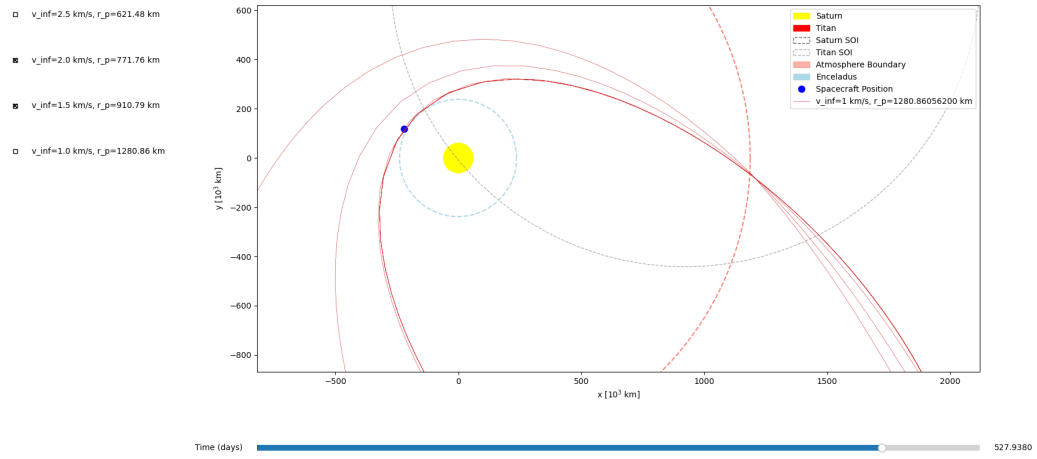


Figure 3.4: Trajectory Example detailed

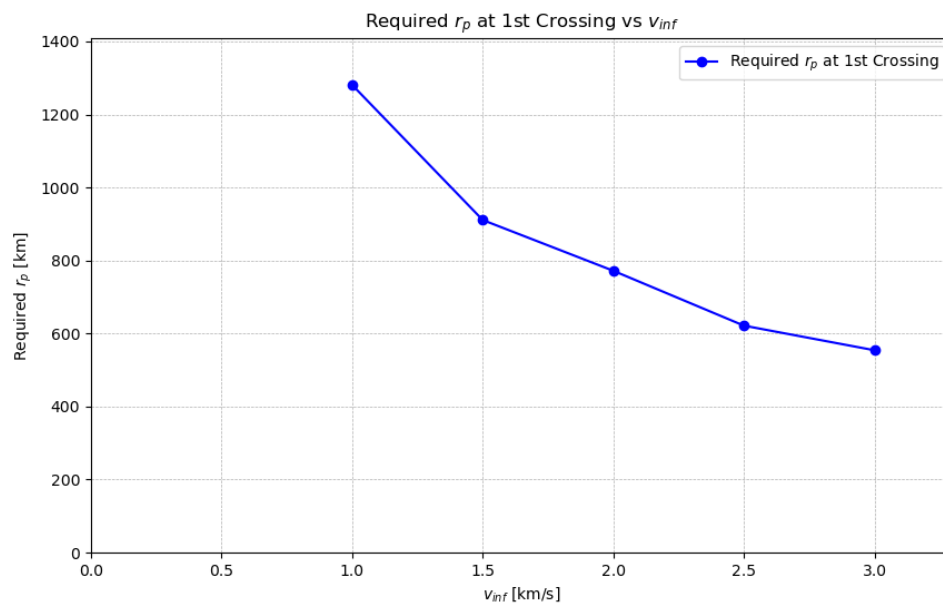


Figure 3.5: Possible First Crossing Distances based on Arrival velocity

3.2 Validation of results

3.2.1 Validation of the atmospheric model

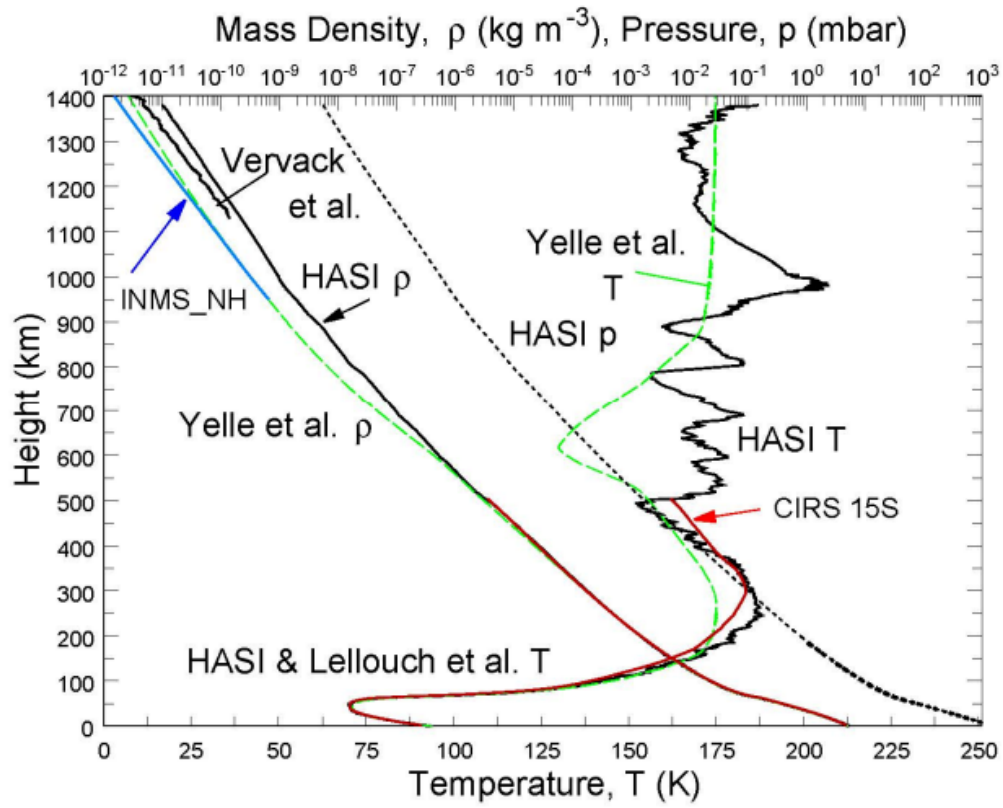


Figure 3.6: Atmospheric Density Data from HASI

3.2.2 Validation of aerobraking results

4 Conclusion

A Appendix Title

A.1 A First Section in the Appendix